

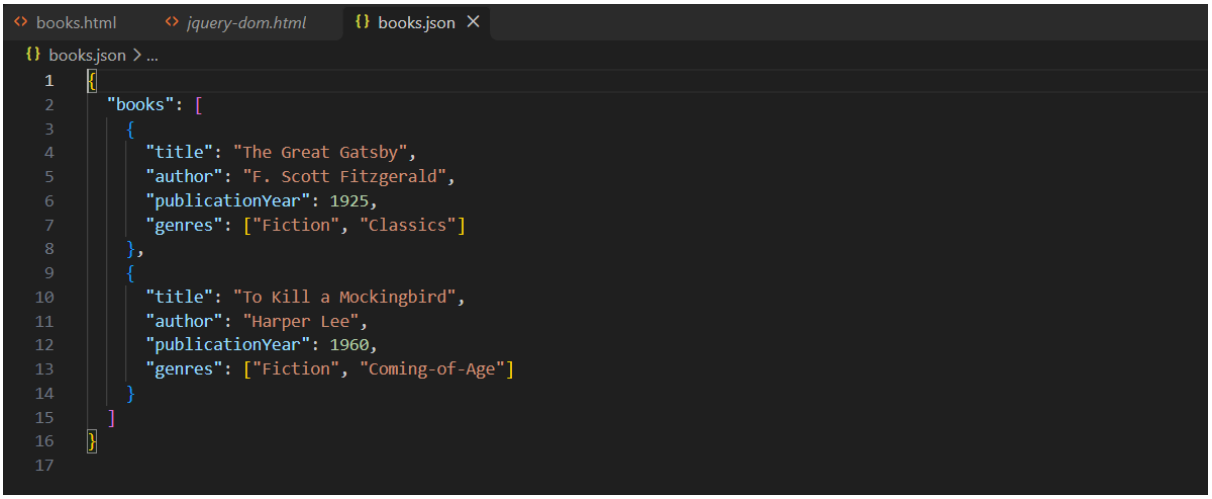

**FACULTY OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY**

# LAB 4B - CLIENT-SIDE SCRIPTING (JSON and jQuery)

**SEMESTER I - SESSION 2026/2027**

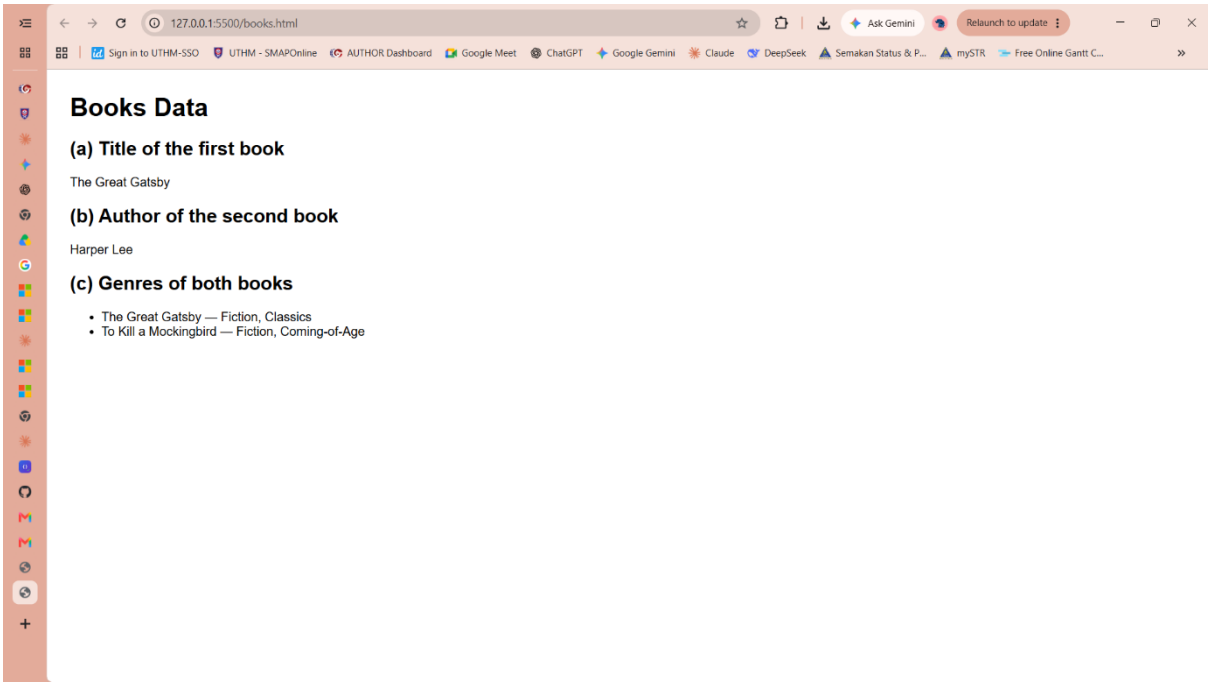
| | |
|---|---|
| COURSE CODE | **BIC21203** |
| COURSE NAME | **WEB DEVELOPMENT** |
| SECTION | **8** |
| LECTURER'S NAME | **DR. CIK FERESA BINTI MOHD FOOZY** |
| STUDENT NAME | **PIRRAVIIN A/L MANIMARAN** |
| MATRIX NO | **CI240058** |

## Question 1 - Create and Structure JSON Data



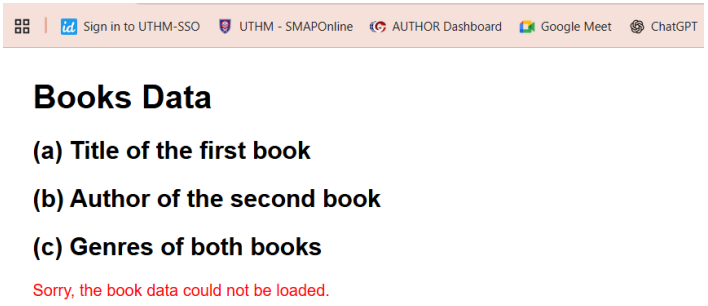

books.json contains a top-level "books" array with two objects. Each object has title and author (strings), publicationYear (number) and genres (array of strings). All property names use double quotes and there are no trailing commas. The file was parsed successfully (Python json.load and the browser's response.json() both accept it).

## Question 2 - Retrieve and Display JSON Data

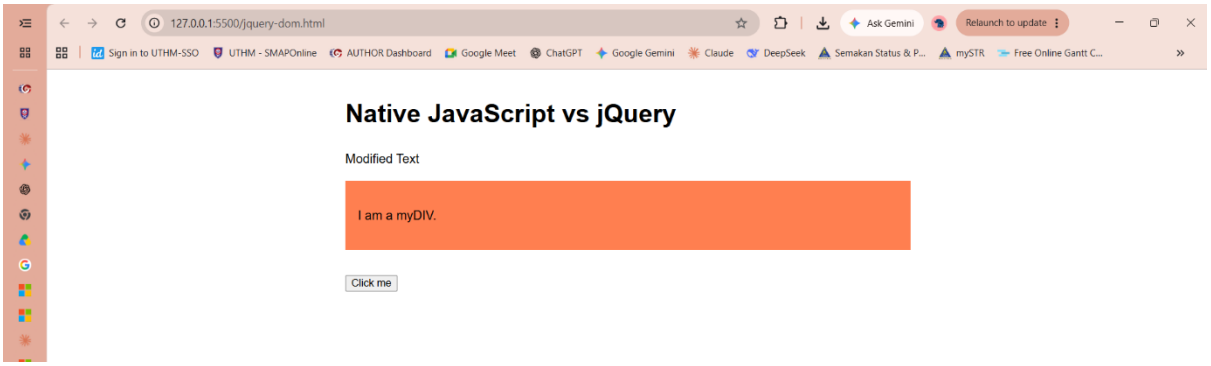


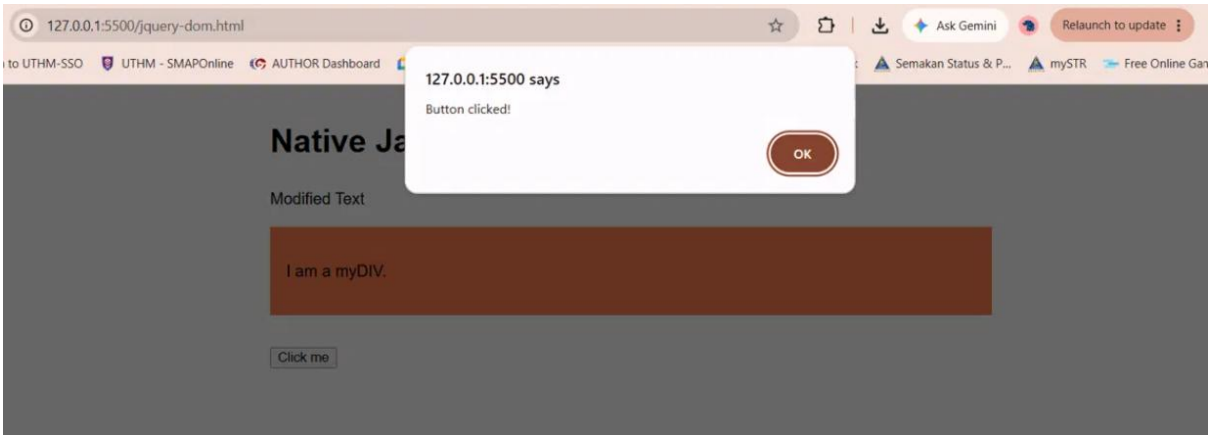

books.html uses fetch("books.json") inside an async function to request the file, and response.json() to convert it into a JavaScript object. The first book's title (a), the second book's author (b) and the genres of both books grouped by title (c) are written into the page using textContent and createElement(), so the data is displayed as plain text. A try/catch block shows an error message on the page if the file cannot be loaded or is not valid JSON. The page was opened through VS Code Live Server (127.0.0.1:5500), not file://.

Error handling test:



## Question 3 - Compare Native JavaScript and jQuery




**Native vs JQuery:**

| | |
|---|---|
| document.getElementById("myElement").textContent = "Modified Text"; | $("#myElement").text("Modified Text"); |
| document.querySelector("#myDIV").classList.add("myStyle"); | $("#myDIV").addClass("myStyle"); |
| document.getElementById("myButton").addEventListener("click", () => alert("Button clicked!")); | $("#myButton").on("click", function () { alert("Button clicked!"); }); |

**Advantage of native JavaScript**: it needs no extra library, so pages load faster and have fewer dependencies to maintain; modern browser APIs (querySelector, classList, fetch) now cover most of what jQuery was used for.

**Situation where jQuery may still be used:** an existing (legacy) project or a plugin-heavy site (e.g. older themes or CMS templates) that already depends on jQuery, where rewriting everything would be costly and risky.

**GitHub repository**: https://github.com/pirraviinuthm/bic21203-lab4b

**- End of Report -**